

Nmap Scan Report

Network Security & Scanning Internship – Task 2

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Tool Used: Nmap 7.98

Target IP Address: 192.168.56.103

Environment: Kali Linux → Metasploitable2 Virtual Machine

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Executive Summary

A network scan was performed against the target host (192.168.56.103) using Nmap. The objective was to identify live hosts, discover open ports, enumerate running services, determine service versions, and identify the operating system running on the target machine.

The scan revealed multiple exposed services including FTP, SSH, Telnet, HTTP, SMB, MySQL, PostgreSQL, and VNC. Service version detection confirmed that the target is running several outdated services commonly found in vulnerable laboratory environments such as Metasploitable2.

Host Discovery

Command

```
ping -c 4 192.168.56.103
```

Result

The target host responded successfully to ICMP Echo Requests.

- Host Status: Online
 - Packet Loss: 0%
 - Average Response Time: < 1 ms
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TCP SYN Scan

Command

```
nmap -sS 192.168.56.103
```

Purpose

To identify open TCP ports and exposed services.

Open Ports Identified

Port	Protocol	Service
21	TCP	FTP
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	TCP	DNS
80	TCP	HTTP
111	TCP	RPCBind
139	TCP	NetBIOS
445	TCP	SMB
512	TCP	Exec
513	TCP	Login
514	TCP	Shell
1099	TCP	Java RMI
1524	TCP	Ingreslock
2049	TCP	NFS
2121	TCP	FTP
3306	TCP	MySQL
5432	TCP	PostgreSQL
5900	TCP	VNC

Port	Protocol	Service
8180	TCP	HTTP

Service Version Detection

Command

```
sudo nmap -sV 192.168.56.103
```

Findings

Service	Version
FTP	vsFTPD 2.3.4
SSH	OpenSSH 4.7p1 Debian
SMTP	Postfix SMTPD
DNS	ISC BIND 9.4.2
HTTP	Apache 2.2.8
SMB	Samba 3.x – 4.x
MySQL	MySQL 5.0.51a
PostgreSQL	PostgreSQL 8.3.x
VNC	Protocol 3.3

Observation

Several services are outdated and known to contain publicly documented vulnerabilities, making the target suitable for security testing and educational purposes.

UDP Scan

Command

```
sudo nmap -sU --top-ports 20 192.168.56.103
```

Findings

Port	State	Service
53	Open	DNS
137	Open	NetBIOS Name Service

Additional ports appeared as Open|Filtered.

Operating System Detection

Command

```
sudo nmap -O 192.168.56.103
```

Results

- Device Type: General Purpose
 - Operating System Family: Linux
 - Kernel Version Range: Linux 2.6.x
 - Network Distance: 1 Hop
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Security Assessment

The target system exposes a large number of network services. Multiple legacy services were identified, including Telnet, FTP, and older versions of Apache, Samba, and MySQL.

Potential risks include:

- Unencrypted remote access protocols
 - Legacy software versions
 - Increased attack surface due to multiple open ports
 - Potential privilege escalation opportunities
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Conclusion

The Nmap assessment successfully identified active services, service versions, open TCP/UDP ports, and operating system details. The target machine presents a large attack surface and serves as an ideal environment for cybersecurity learning and vulnerability assessment exercises.

Future work will include:

- Vulnerability Scanning using OpenVAS/Nessus
- Firewall Rule Configuration using iptables
- Risk Analysis and Remediation Recommendations